

Mount Kenya University



School of Computing and Informatics
BIT3208: ADVANCED WEB DESIGN AND DEVELOPMENT
PROJECT LOGBOOK

Student Name: Isaac Kiarie Chege

Admission Number: BBIT/2024/54601

Course: Bachelor of Business Information Technology

Unit Code: Advanced Web Design and Development

Lecture Name: Michael Nyoro

Semester : 3.2

Technologies Used: PHP , MySQL, HTML, CSS , Javascript and Xampp

Github Repository: <https://github.com/Isaac-kiarie/Ecommerce-Website-Application.git>

TABLE OF CONTENTS

Week 1

Environment setup

Fig 1: Installation of XAMPP/WAMP/Laragon	3
Fig 2: Apache and MySQL Running	4
Fig 3: Localhost Test Page.....	4
Fig 4: Hello World Test.....	5
Fig 5: Database Connection Test.....	5

Week 2

Wireframes and GUI designs

Fig 1: Hand-drawn Wireframe	6
Fig 2: Dashboard Layout Design.....	7
Fig 3: Mobile View Mockup.....	8
Fig 4: Color Theme and Navigation Structure.....	8
Fig 5: GUI Prototype in Figma/Canva/Draw.io.....	9

Week 3

JavaScript and PHP basics

Fig 1: JavaScript Form Validation.....	10
Fig 2: Password Strength Checker.....	11
Fig 3: PHP Syntax Practice.....	12
Fig 4: Database Connection Script.....	13
Fig 5: Dynamic User Input Handling.....	14

Week 4

Database Design and CRUD Operations

Login form screenshots.....	15
Form validation screenshots.....	16
PHP processing screenshots.....	17
Folder structure screenshots.....	17
Database connection screenshots.....	18
Code snippets.....	18
GitHub updates.....	19

Week 5

Database screenshots.....	20
Table creation screenshots.....	21
CRUD operation screenshots.....	21
Login database integration screenshots.....	21
SQL query screenshots.....	21
GitHub updates.....	22

WEEK 1: Local Environment Setup

Fig 1.1: XAMPP Installation Success

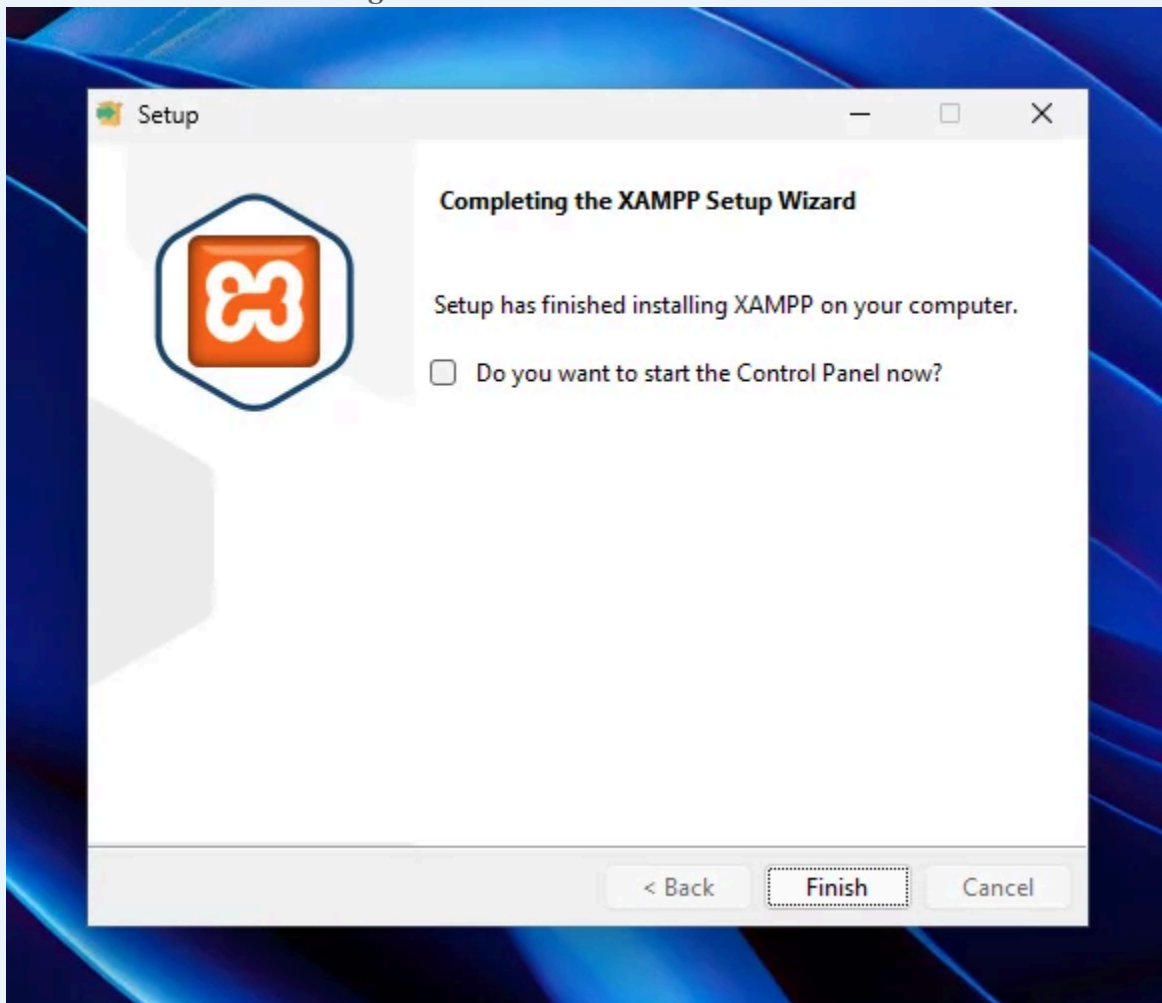


Fig 2: Apache and MySQL Running

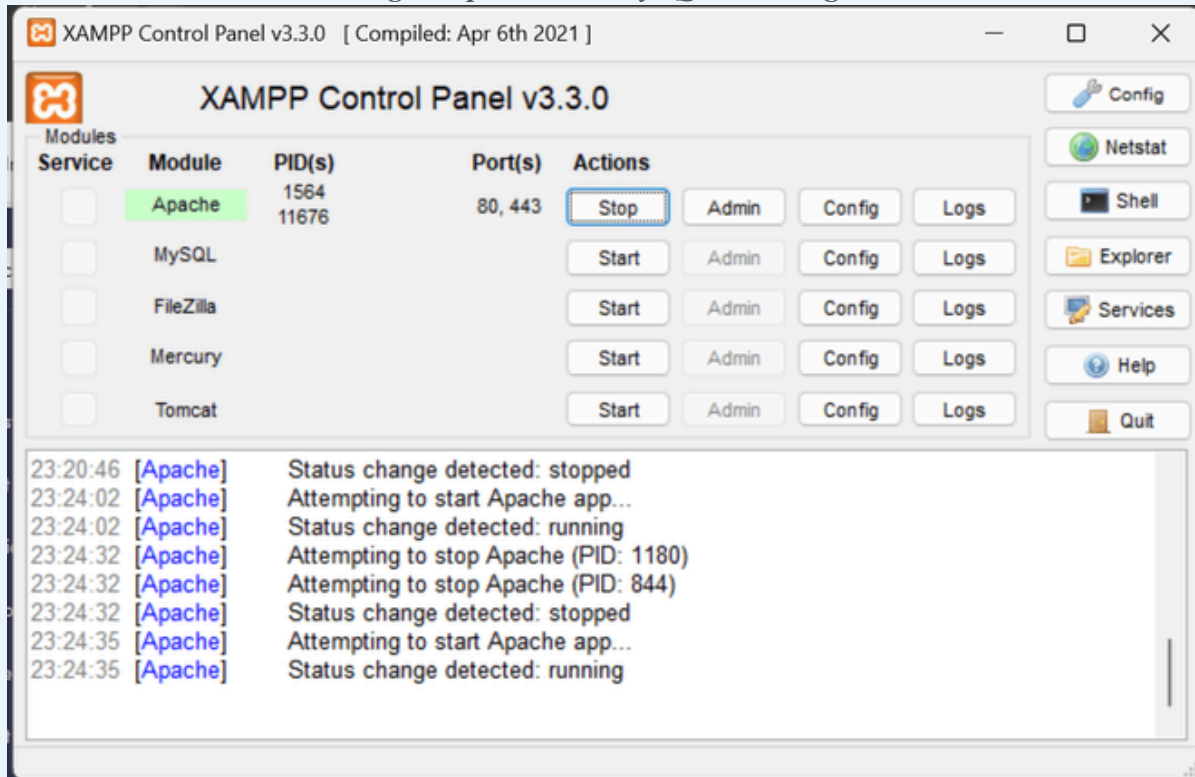


Fig 3: Localhost Test Page

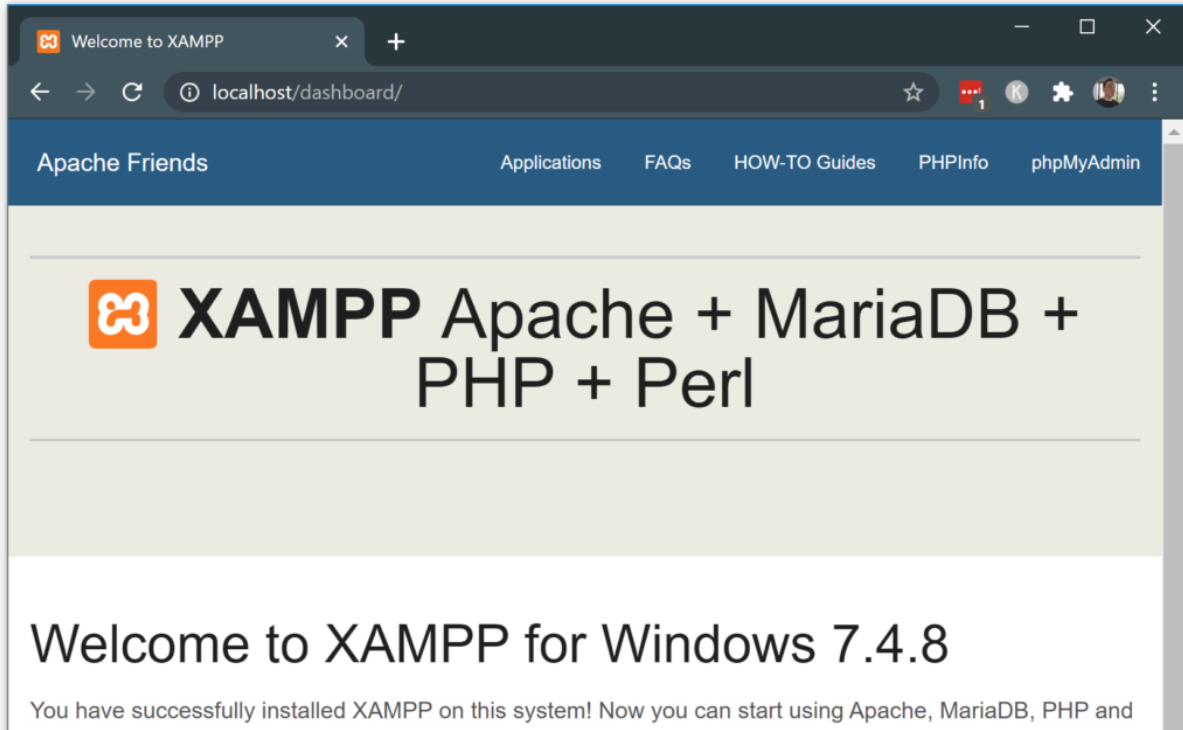
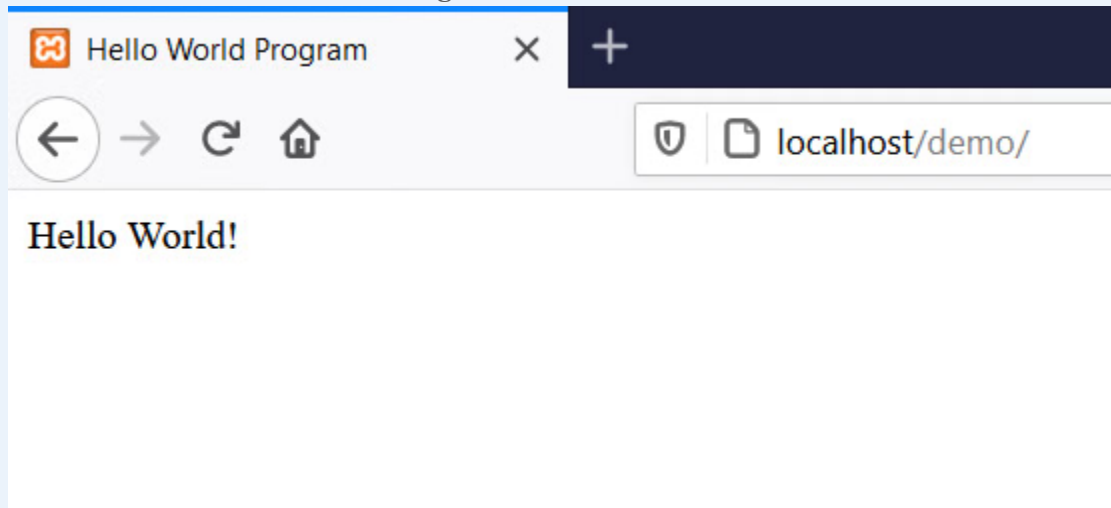


Fig 4: Hello World Test



http://localhost/ecommerce-app/test.php displaying version info and Hello World output.

Fig 5: Database Connection Test



Student Reflection

This week I set up XAMPP on Windows and confirmed that Apache and MySQL services run simultaneously which is essential for a PHP–MySQL application. I also did a hello world test to check if it was functional, the database connection worked well using php and xampp it was easier to connect to it, just a few code it was do

WEEK 2: Wireframe and GUI Design

Fig 1: Hand-drawn Wireframe



Fig 2: Dashboard Layout Design

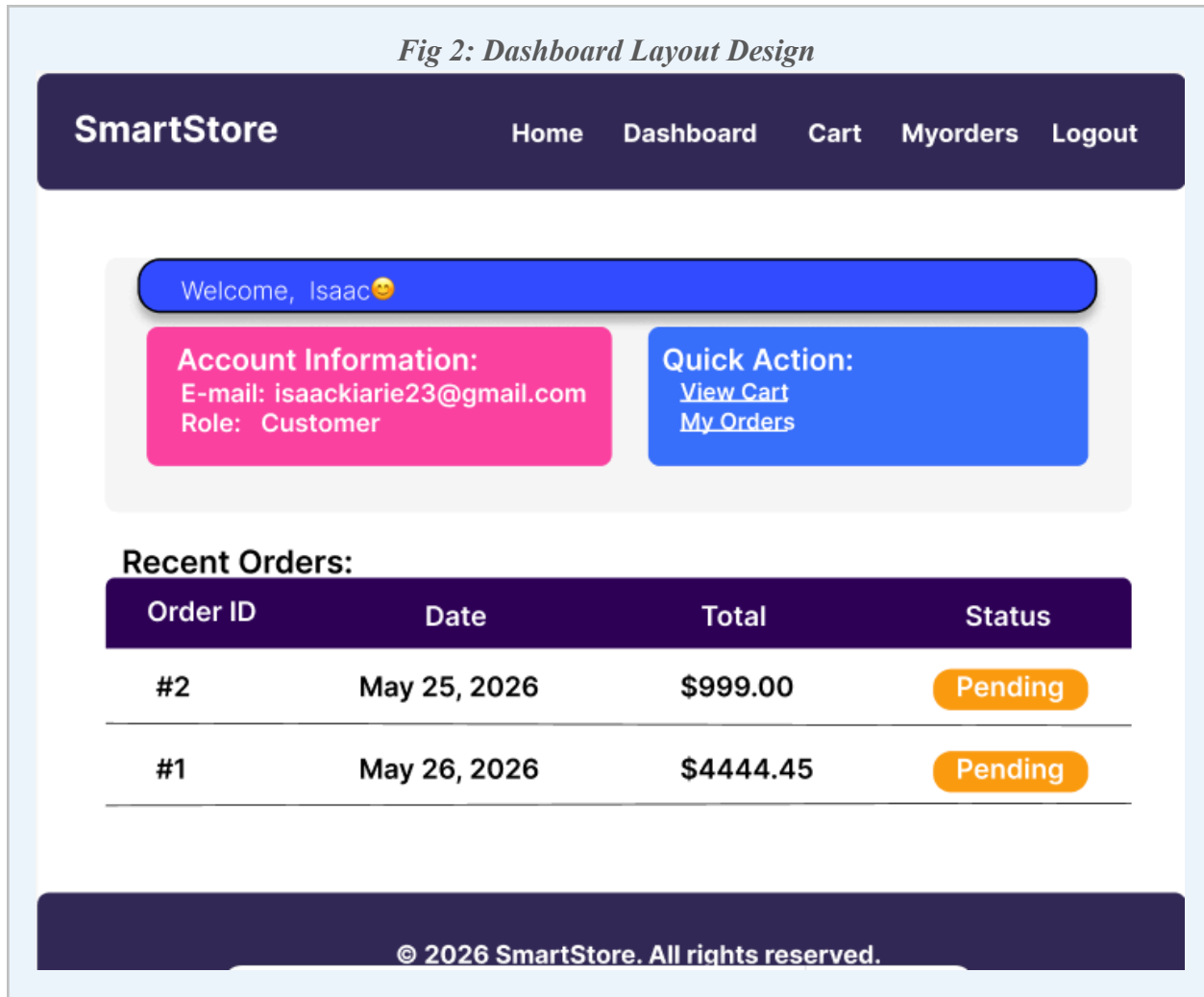


Fig 3: Mobile View Mockup

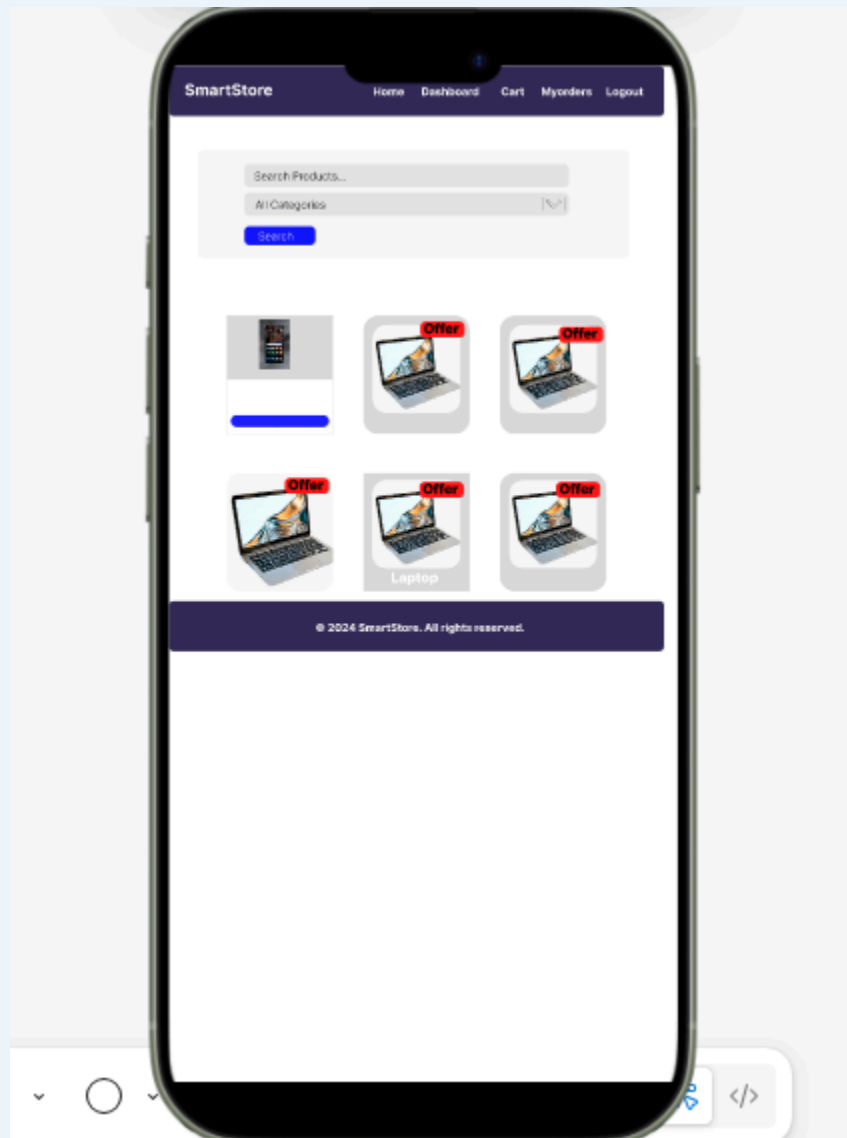
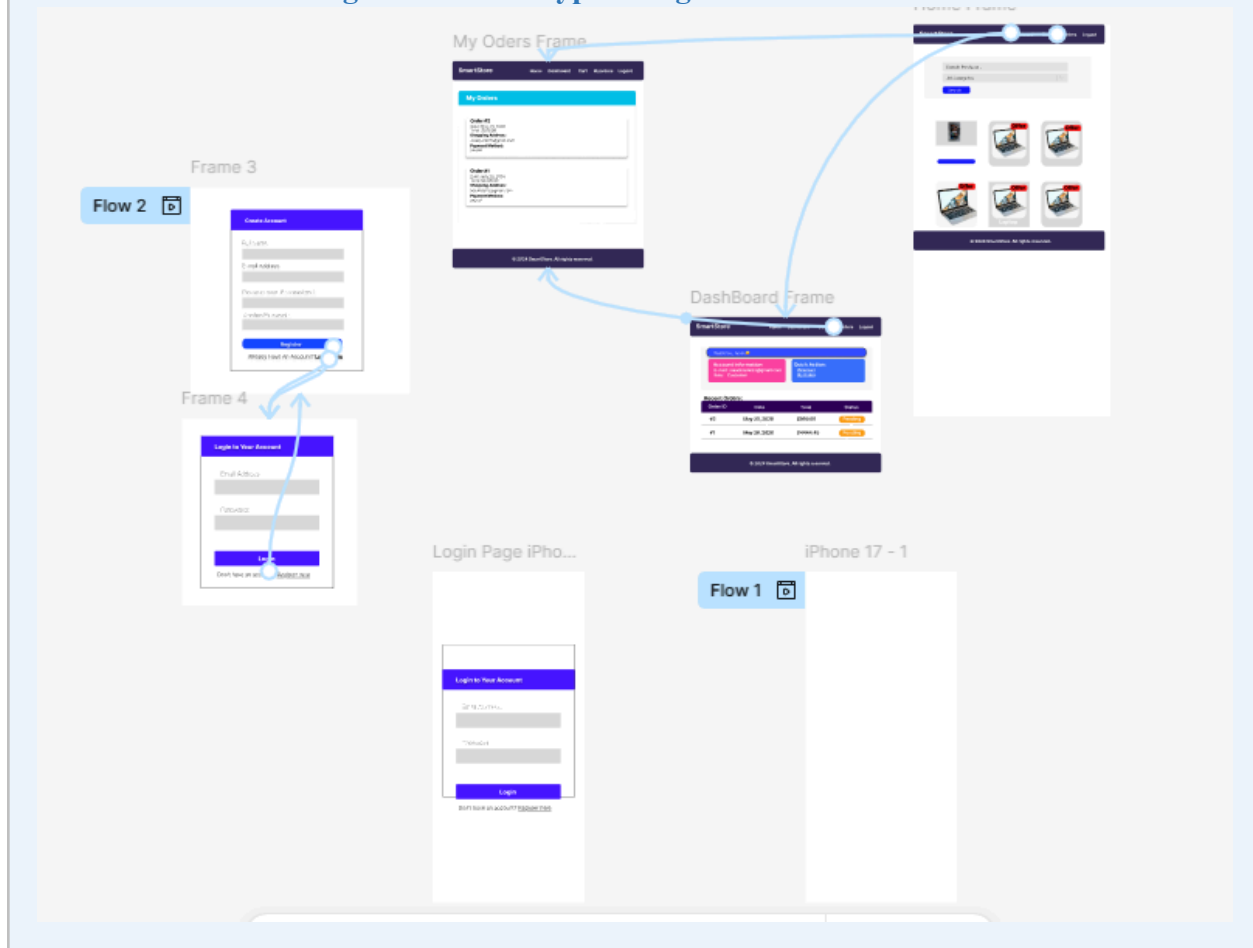


Fig 4: Color Theme and Navigation Structure

SmartStore

Home Login Register

Fig 5: GUI Prototype in Figma/Canva/Draw.io



Student Reflection

Within week two i worked on Hand-drawn Wireframe, Dashboard Layout Design Mobile View Mockup,Color Theme and Navigation Structure GUI Prototype in Figma/Canva/[Draw.io](#).

It was awesome and a bit challenging on the mobile mock-up.

I understood Have to use different tools of the figma application used for making the wireframes.

WEEK 3: Javascript and PHP Basics

Fig 1: JavaScript Form Validation

The screenshot shows a web browser window displaying the 'EcommerceStore' registration page. The page has a dark blue header with the store name and navigation links for Home, Login, and Register. The main content area features a 'Create Account' form with a blue title bar. The form includes a red error message at the top: 'Password must be at least 6 characters'. Below this, there are four input fields: 'Full Name' (containing 'John'), 'Email Address' (containing 'JohnMark12@gmail.com'), 'Password (min 6 characters)' (empty), and 'Confirm Password' (empty). The browser's taskbar at the bottom shows various application icons and the system clock indicating 09:51.

EcommerceStore Home Login Register

Create Account

Password must be at least 6 characters

Full Name
John

Email Address
JohnMark12@gmail.com

Password (min 6 characters)

Confirm Password

Fig 2: Password Strength Checker

Password

Create a strong password

Password is required

Password Strength: undefined

☐ ✓ At least 8 characters

☐ ✓ At least 1 uppercase letter (A-Z)

☐ ✓ At least 1 lowercase letter (a-z)

☐ ✓ At least 1 number (0-9)

☐ ✓ At least 1 special character (!@#\$%^&*)

☒ ✓ Not a common password

 **Tips for a strong password:**

- Use a mix of uppercase, lowercase, numbers, and symbols
- Make it at least 12 characters for better security

CRUD OPERATION

SmartStoreHomeDashboardCart (2)My OrdersLogout

Shopping Cart

Product	Price	Quantity	Subtotal	Action
T-Shirt Stock: 100 available	\$19.99	1 <button>Update</button>	\$19.99	<button>Remove</button> <button>Save for Later</button>
Novel Stock: 75 available	\$14.99	1 <button>Update</button>	\$14.99	<button>Remove</button> <button>Save for Later</button>

Order Summary
Subtotal (2 items): **\$34.98**
Shipping: Calculated at checkout
Total: **\$34.98**

Proceed to Checkout →Continue Shopping

Fig 3: PHP Syntax Practice

```
1  <?php
2  echo "<!DOCTYPE html>";
3  echo "<html>";
4  echo "<head>";
5  echo "<title>PHP Test</title>";
6  echo "<style>
7      body {
8          font-family: Arial, sans-serif;
9          background: #f4f4f4;
10         text-align: center;
11         padding-top: 100px;
12     }
13     .card {
14         background: white;
15         width: 400px;
16         margin: auto;
17         padding: 30px;
18         border-radius: 10px;
19         box-shadow: 0 0 10px rgba(0,0,0,0.1);
20     }
```

Fig 4: Database Connection Script

```
class Database {
    private $host = "localhost";
    private $db_name = "ecommerce_db";
    private $username = "root";
    private $password = "";
    public $conn;

    public function getConnection() {
        $this->conn = null;

        try {
            $this->conn = new PDO(
                "mysql:host=" . $this->host . ";dbname=" . $this->db_name,
                $this->username,
                $this->password,
                array(
                    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
                )
            );
        } catch (PDOException $e) {
            echo "Database error: " . $e->getMessage();
        }
    }
}
```

Fig 5: Dynamic User Input Handling

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password

👁️

Password is required

Password Strength: undefined

☐ ✓ At least 8 characters

☐ At least 8 characters and 1 number

☐ At least 8 characters, 1 number, and 1 special character

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password

Password is required

Progress bar showing strength

Password Strength: undefined

☐ ✓ At least 8 characters

☐ At least 12 characters

☐ At least 16 characters

☐ At least 20 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password

👁️

Password is required

Password Strength: undefined

✓ At least 8 characters

[illegible]

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password



Password is required

Progress bar showing strength level

Password Strength: undefined

☒ ✓ At least 8 characters

☐ ✗ Not strong enough

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password



Password is required

Password Strength: undefined

☐

✓ At least 8 characters

☐

At least 8 characters and 1 number

☐

At least 8 characters, 1 number, and 1 special character

☐

At least 8 characters, 1 number, 1 special character, and 1 capital letter

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Student Reflection

Within week 3 i worked on, JavaScript Form Validation, Password Strength Checker, PHP Syntax Practice, Database Connection Script, Dynamic User Input Handling. My views on this is that I have seen the importance of having a valid credentials from the user and also the ability of the system to indicate to the user the places He/She has gone a miss.

Student Reflection

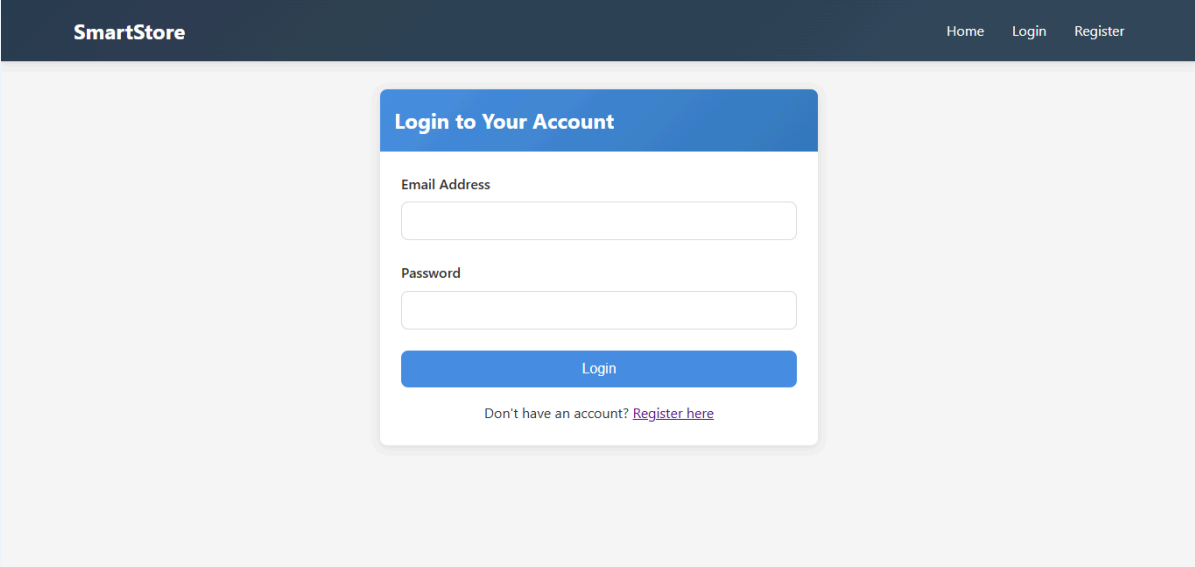
Within week 3 i worked on, JavaScript Form Validation, Password Strength Checker, PHP Syntax Practice, Database Connection Script, Dynamic User Input Handling. My views on this is that I have seen the importance of having a valid credentials from the user and also the ability of the system to indicate to the user the places He/She has gone a miss.

Student Reflection

Within week 3 i worked on, JavaScript Form Validation, Password Strength Checker, PHP Syntax Practice, Database Connection Script, Dynamic User Input Handling. My views on this is that I have seen the importance of having a valid credentials from the user and also the ability of the system to indicate to the user the places He/She has gone a miss.

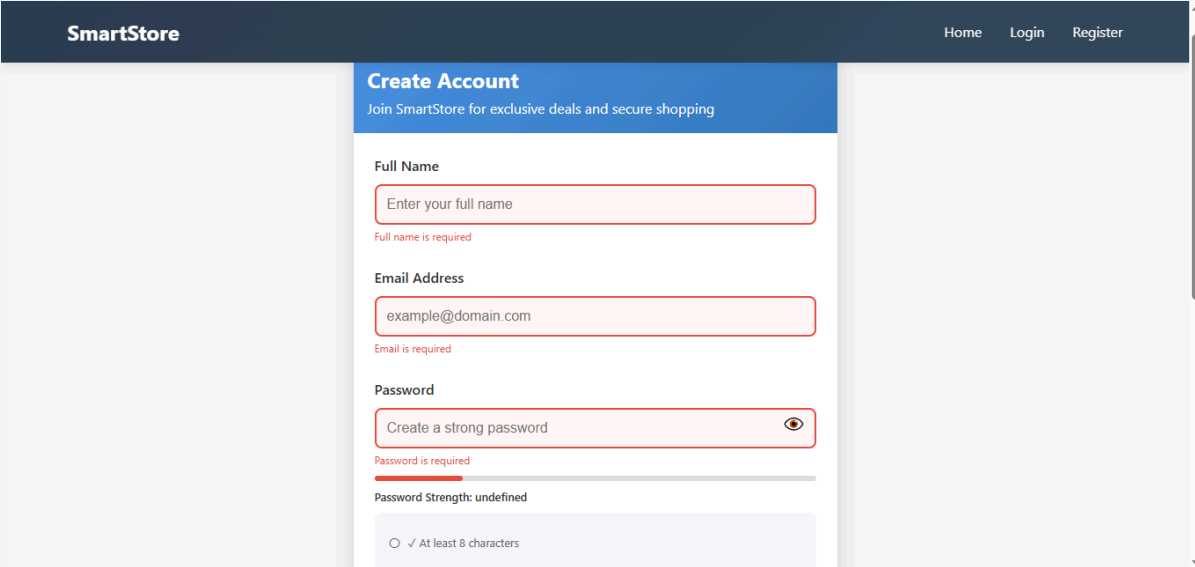
WEEK 4: Database Design and CRUD Operations

Login form screenshots



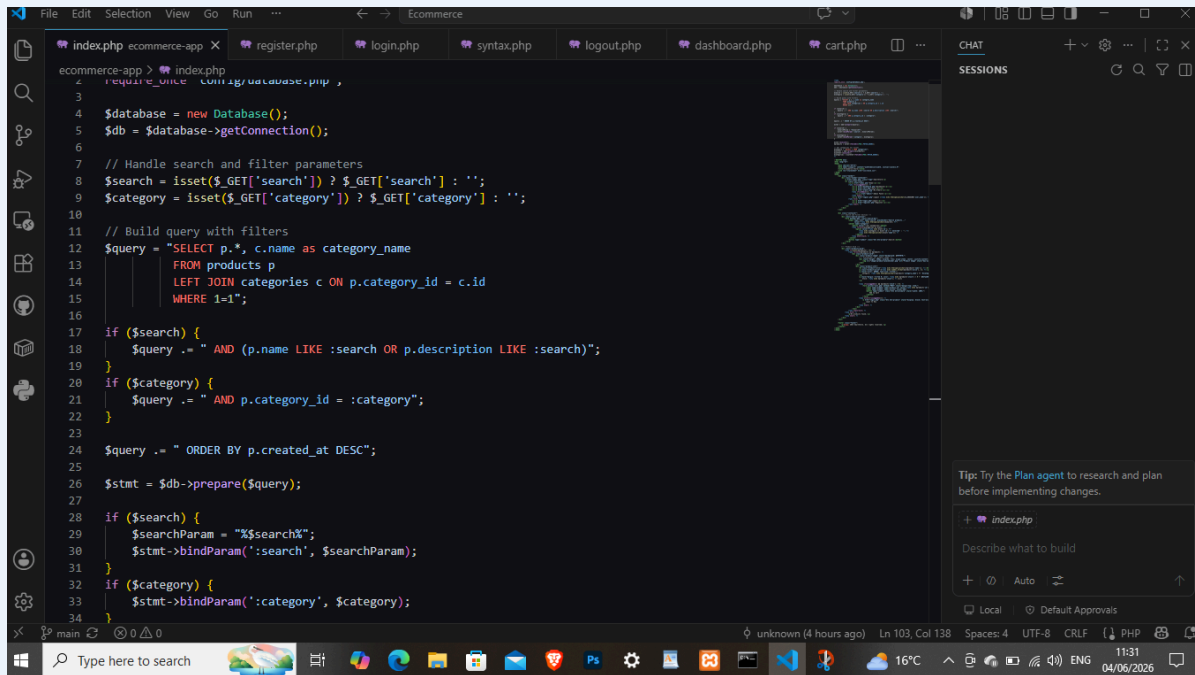
The screenshot shows the login form for SmartStore. The header is dark blue with the SmartStore logo on the left and links for Home, Login, and Register on the right. The main content area is light gray. The login form is a white card with a blue header that says "Login to Your Account". It contains two input fields: "Email Address" and "Password". Below the password field is a blue "Login" button. At the bottom of the card, there is a link that says "Don't have an account? [Register here](#)".

Form validation screenshots



The screenshot shows the "Create Account" form for SmartStore. The header is dark blue with the SmartStore logo on the left and links for Home, Login, and Register on the right. The main content area is light gray. The form is a white card with a blue header that says "Create Account" and a sub-header "Join SmartStore for exclusive deals and secure shopping". It contains three input fields: "Full Name", "Email Address", and "Password". The "Full Name" field has a red border and a red error message "Full name is required". The "Email Address" field has a red border and a red error message "Email is required". The "Password" field has a red border and a red error message "Password is required". Below the password field is a "Password Strength" section with a red progress bar and a radio button labeled "At least 8 characters".

PHP Processing screenshots

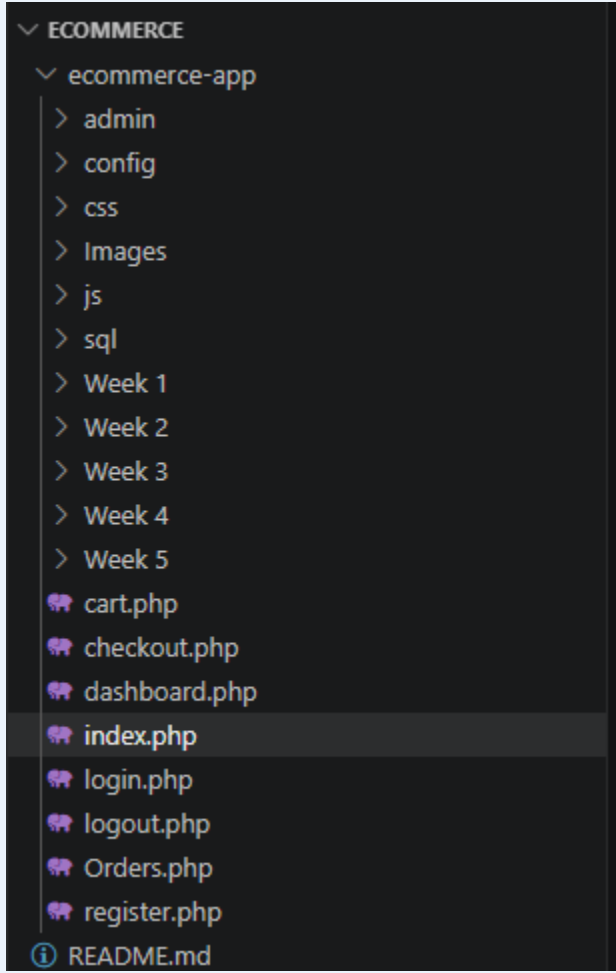


```
File Edit Selection View Go Run ... Ecommerce
index.php ecommerce-app X register.php login.php syntax.php logout.php dashboard.php cart.php ... CHAT
SESSIONS

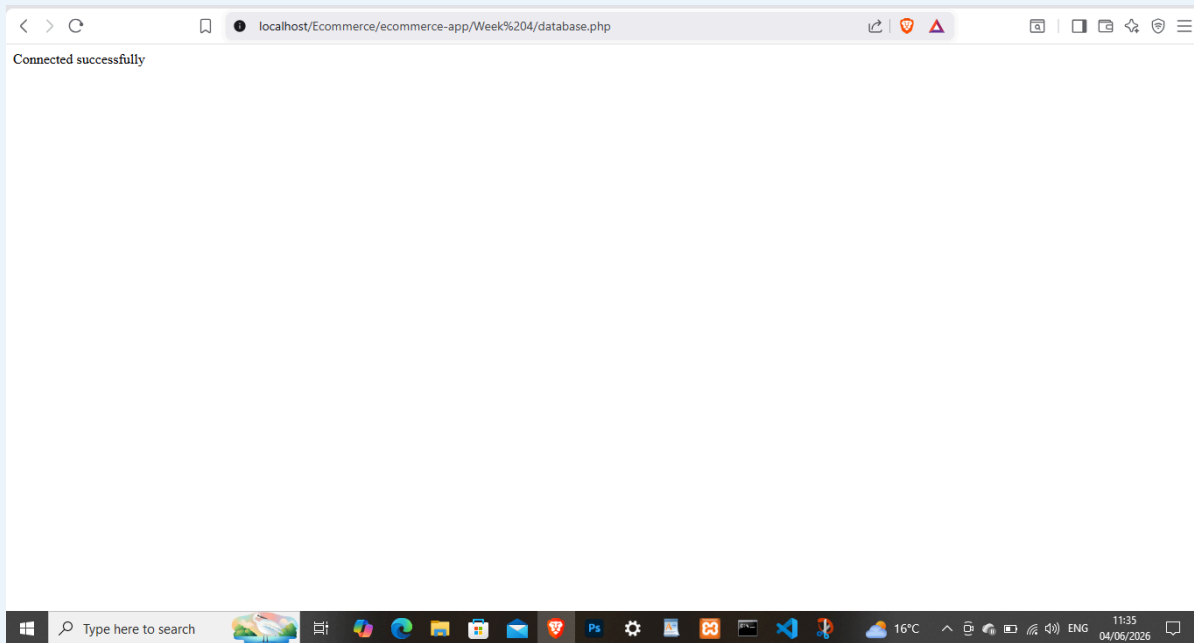
ecommerce-app > index.php
1 require_once 'config/database.php';
2
3
4 $database = new Database();
5 $db = $database->getConnection();
6
7 // Handle search and filter parameters
8 $search = isset($_GET['search']) ? $_GET['search'] : '';
9 $category = isset($_GET['category']) ? $_GET['category'] : '';
10
11 // Build query with filters
12 $query = "SELECT p.*, c.name as category_name
13           FROM products p
14           LEFT JOIN categories c ON p.category_id = c.id
15           WHERE 1=1";
16
17 if ($search) {
18     $query .= " AND (p.name LIKE :search OR p.description LIKE :search)";
19 }
20 if ($category) {
21     $query .= " AND p.category_id = :category";
22 }
23
24 $query .= " ORDER BY p.created_at DESC";
25
26 $stmt = $db->prepare($query);
27
28 if ($search) {
29     $searchParam = "%$search%";
30     $stmt->bindParam(':search', $searchParam);
31 }
32 if ($category) {
33     $stmt->bindParam(':category', $category);
34 }

Tip: Try the Plan agent to research and plan before implementing changes.
+ index.php
Describe what to build
+ Auto
Local Default Approvals
unknown (4 hours ago) Ln 103, Col 138 Spaces: 4 UTF-8 CRLF PHP
16°C 11:31 04/06/2026
```

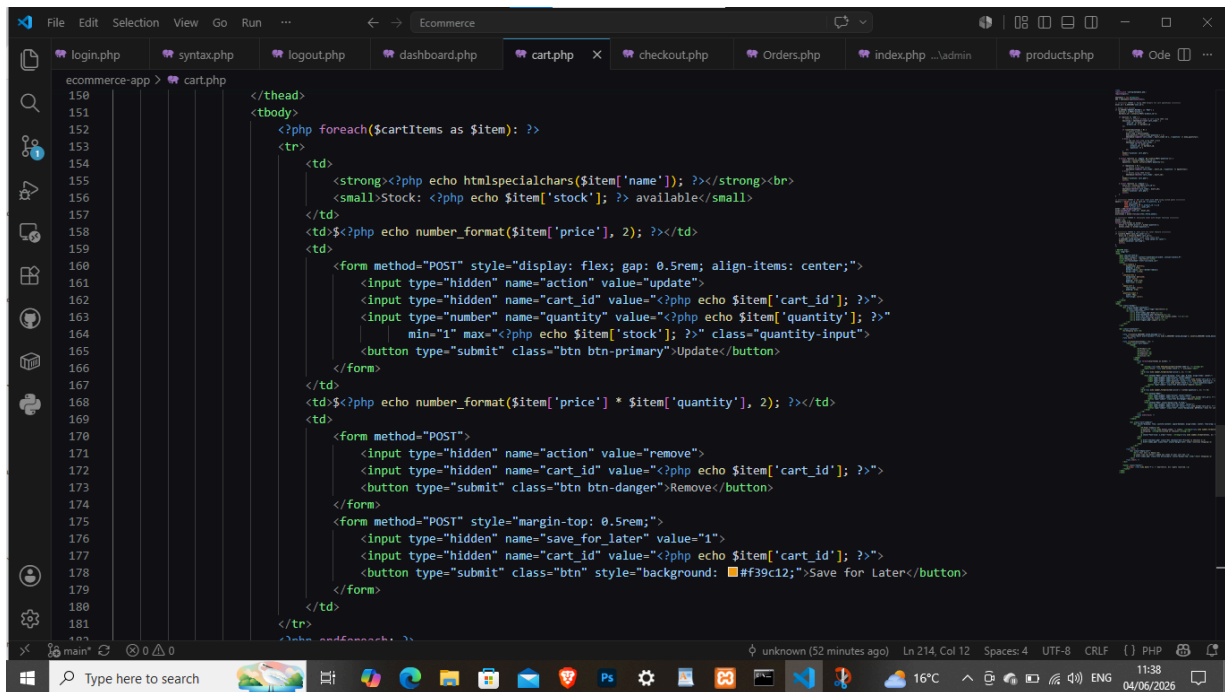
Folder structure screenshots



Database connection screenshots



Code snippets



GitHub updates

The screenshot shows a web browser displaying the GitHub repository page for 'Isaac-kiarie/Ecommerce-Website-Application'. The URL in the address bar is 'github.com/Isaac-kiarie/Ecommerce-Website-Application/tree/main/ecommerce-app'. The left sidebar shows the file structure with folders for 'Week 1', 'Week 3', 'admin', 'config', 'css', and 'js'. The main content area displays a table of files and folders with their commit messages and dates.

Name	Last commit message	Last commit date
..		
Week 1	This is a second commit for the weeks	1 hour ago
Week 3	This is week 3 Syntax for php	51 minutes ago
admin	This is a second commit for the weeks	1 hour ago
config	This is a second commit for the weeks	1 hour ago
css	This is a second commit for the weeks	1 hour ago
js	first commit	4 hours ago
Orders.php	This is a second commit for the weeks	1 hour ago
cart.php	This is a second commit for the weeks	1 hour ago
checkout.php	This is a second commit for the weeks	1 hour ago
dashboard.php	This is a second commit for the weeks	1 hour ago
index.php	first commit	4 hours ago
login.php	first commit	4 hours ago
logout.php	first commit	4 hours ago

Student Reflection

.Within this week i got my hands dirty on the code making the Validation forms ensuring all the necessary information has been imputed by the user, I have understood about, backend processing, Process forms using PHP, request-response workflow, login logic, sessions basics, Structure projects professionally. I find it easier now with the concepts that used to seem complicated.

WEEK 5: User Authentication and Session Management

Database screenshots

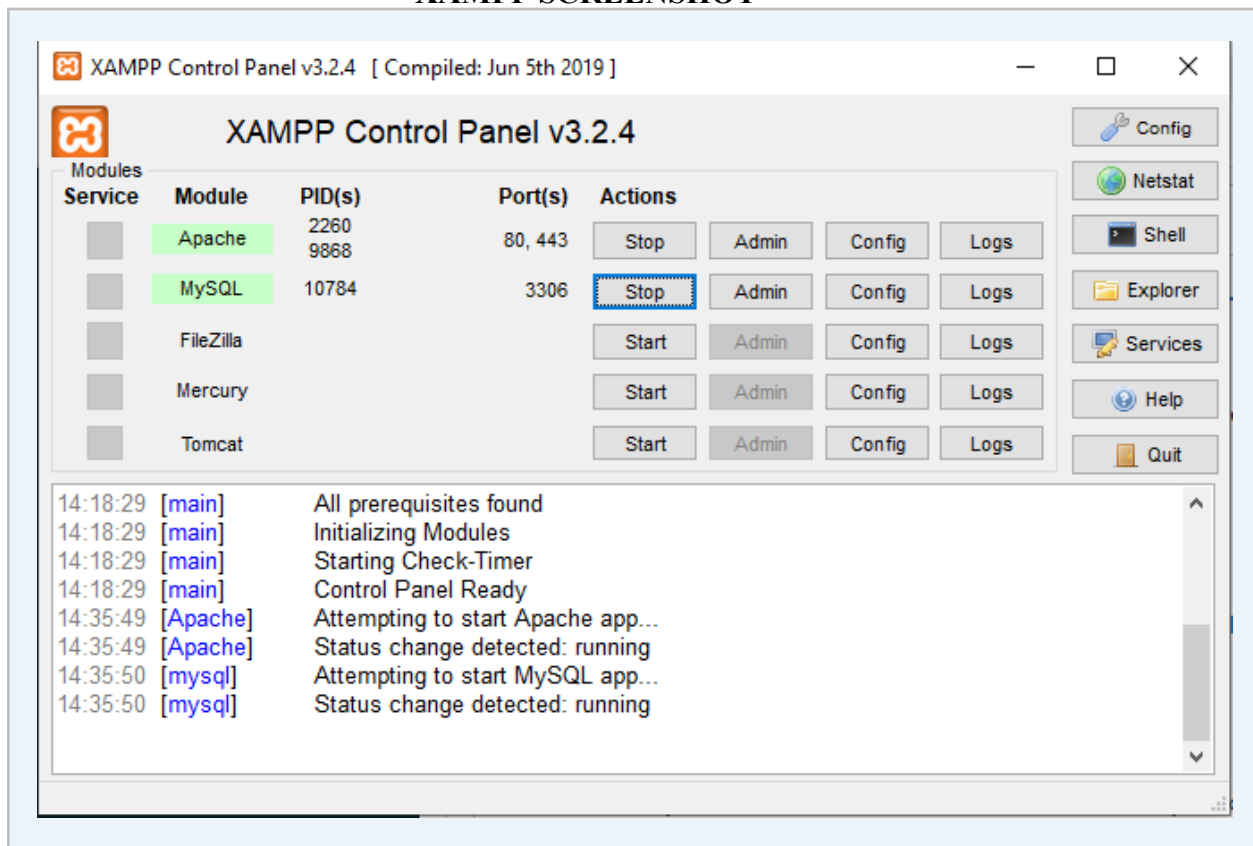
The screenshot shows the phpMyAdmin interface for a database named 'ecommerce_db'. The left sidebar shows the database structure with tables: cart_items, categories, orders, order_items, products, and users. The main panel displays the 'Structure' tab for the 'ecommerce_db' database. It shows a list of tables with their respective actions (Browse, Structure, Search, Insert, Empty, Drop) and summary statistics (Rows, Type, Collation, Size, Overhead). Below the table list, there is a 'Create new table' button and a form to create a new table with a table name and number of columns.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ cart_items	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	12	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

Table Creation Shots.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ cart_items	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	12	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

XAMPP SCREENSHOT



CRUD operation screenshots

The screenshot shows the 'SmartStore' application's shopping cart interface. The browser address bar indicates the URL is `localhost/Ecommerce/eCommerce-app/cart.php`. The page features a navigation bar with links to Home, Dashboard, Cart (2), My Orders, and Logout. The main content area is titled 'Shopping Cart' and displays a table with the following items:

Product	Price	Quantity	Subtotal	Action
T-Shirt Stock: 100 available	\$19.99	1	\$19.99	Update Remove Save for Later
Novel Stock: 75 available	\$14.99	1	\$14.99	Update Remove Save for Later

Below the cart items, an 'Order Summary' section shows a subtotal of \$34.98 for 2 items, with shipping calculated at checkout, resulting in a total of \$34.98. A green 'Proceed to Checkout' button and a purple 'Continue Shopping' link are provided. The Windows taskbar at the bottom shows the system time as 12:46 on 04/06/2026.

SQL query screenshots

The screenshot shows the phpMyAdmin interface for the 'eCommerce_db' database. The browser address bar shows the URL `localhost/phpmyadmin/index.php?route=/database/sql&db=eCommerce_db`. The left sidebar displays the database structure, including tables like `accessory-shop`, `cart_items`, `categories`, `orders`, `order_items`, `products`, `users`, `information_schema`, `mysql`, `performance_schema`, `phpmyadmin`, and `test`. The main area is the SQL query editor, which contains the following SQL code:

```
41 quantity INT DEFAULT 1,
42 added_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
43 FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
44 FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE,
45 UNIQUE KEY unique_cart_item (user_id, product_id)
46 );
47
48 -- Orders table
49 CREATE TABLE orders (
50 id INT PRIMARY KEY AUTO_INCREMENT,
51 user_id INT,
52 order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
53 total_amount DECIMAL(10,2),
54 status ENUM('pending', 'processing', 'completed', 'cancelled') DEFAULT 'pending',
55 shipping_address TEXT,
56 )
```

The editor includes buttons for 'Clear', 'Format', and 'Get auto-saved query'. Below the query editor, there are checkboxes for 'Bind parameters', 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks'. A 'Go' button is at the bottom right. The Windows taskbar at the bottom shows the system time as 12:49 on 04/06/2026.

GitHub updates

The screenshot shows a web browser displaying the GitHub repository page for 'Isaac-kiarie/Ecommerce-Website-Application'. The browser's address bar shows the URL 'github.com/Isaac-kiarie/Ecommerce-Website-Application/tree/main/eCommerce-app'. The page is divided into two main sections: a file tree on the left and a commit history table on the right.

File Tree (Left):

- main
- ecommerce-app
 - Week 1
 - Apache and MySQL Running.p...
 - HelloWorld.php
 - HelloWorld.png
 - Localhost_Test_Page.png
 - installation_of_Xampp.png
 - Week 3
 - syntax.php
 - admin
 - config
 - css
 - js
 - Orders.php
 - cart.php

Commit History Table (Right):

Name	Last commit message	Last commit date
..		
Week 1	This is a second commit for the weeks	2 hours ago
Week 3	This is week 3 Syntax for php	2 hours ago
admin	This is a second commit for the weeks	2 hours ago
config	This is a second commit for the weeks	2 hours ago
css	This is a second commit for the weeks	2 hours ago
js	first commit	6 hours ago
Orders.php	This is a second commit for the weeks	2 hours ago
cart.php	This is a second commit for the weeks	2 hours ago
checkout.php	This is a second commit for the weeks	2 hours ago
dashboard.php	This is a second commit for the weeks	2 hours ago
index.php	first commit	6 hours ago
login.php	first commit	6 hours ago
logout.php	first commit	6 hours ago

Student Reflection

Within this week i got chance to work on different operations such as Create, Update, Delete and read, i was unable to understand things like retrieving of data and information from the database towards the front of the website, it was exciting.

